

Parham Saremi

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Education

McGill University – Mila

Montreal, Canada

M.Sc. in Electrical and Computer Engineering

2024 – Present

- Supervised by Prof. Tal Arbel – Overall GPA: 4/4

Sharif University of Technology

Tehran, Iran

B.Sc. in Computer Engineering

2018 – 2023

- Overall GPA: 18.70/20 – Overall Major GPA: 19.15/20

Publications

Discovering Latent Graphs with GFlowNets for Diverse Conditional Image Generation *NeurIPS 2025*

B. Trang, **P. Saremi**, ..., T. Arbel, L. Fei-Fei, E. Adeli,

RL4Med-DDPO: Reinforcement Learning for Controlled Guidance Towards Diverse Medical Image Generation using Vision-Language Foundation Models *MICCAI 2025*

P. Saremi*, A. Kumar*, M. Mohammed, Z. TehraniNasab., T. Arbel

Conditional Diffusion Models are Medical Image Classifiers that Provide Explainability and Uncertainty for Free *MIDL 2025*

P. Saremi*, G. M. Favero*, E. Kaczmarek, B. Nichyporuk, T. Arbel

Towards Reliable Human Pose Forecasting with Uncertainty *IEEE RAL*

S. Saadatnejad, **P. Saremi***, M. Mirmohammadi*, M. Daghyani*, ..., T. Mordan, A. Alahi

Reconstruction of 3D Interaction Models from Images Using Shape Prior *ICCV R6D Workshop*

M. Mirmohammadi, **P. Saremi**, Y.-L. Kuo, X. Wang

Experience

Mila – Quebec Artificial Intelligence Institute

Montreal, Canada

Graduate Research Student

May 2024 – Present

- Conducting research in **medical imaging** and **generative modeling** under the supervision of Prof. Tal Arbel.
- Working on **Diffusion Models** for explainability and uncertainty quantification in medical images.
- Trained **stable-diffusion** on medical data and fine-tuned using **policy optimization** to improve alignment by 11%

ETH Zürich – AIT Lab

Zurich, Switzerland (Remote)

Research Intern

Jul 2022 – Jul 2023

- Designed and implemented pipelines to generate **3D human and object models** from **single images** while reasoning about their interaction.
- Used the decoder of a pre-trained **VQ-VAE** model to generate object meshes from images.

EPFL – VITA Lab

Lausanne, Switzerland (Remote)

Research Intern

Dec 2021 – Jul 2022

- Developed and maintained **UnPOSed**, an **open-source** toolbox for **forecasting** a sequence of human pose in future.
- Designed and evaluated a method for human motion prediction that **improved the results of various SOTA models** on multiple datasets from **2% to 5%**.

University California Irvine & Sharif University of Technology

Tehran, Iran

Research Collaborator – B.Sc. Project

Nov 2021 – Jul 2022

- Worked with **vision-language models** on a joint research project between SUT and UCI for my BSc thesis.
- Designed and developed a multi-modal model using **GNNs and Transformers** to predict cuisine using ingredient information and recipes.

Skills

- **ML Skills:** Python, Pytorch, Pytorch3D, Numpy, Pandas, Sklearn, Matplotlib, HuggingFace, CometML
- **Development & Tools:** Linux, Git, GitHub, L^AT_EX, Excel, CI/CD, Code Review
- **Programming Languages:** Python, F#, C, C++, Java, C#, MIPS, X86, R